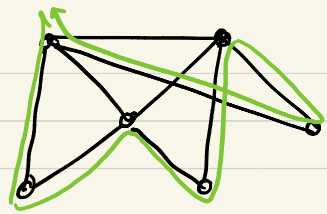
Hamiltonian Cycle: a cycle that visits every verdex exactly once.



Hamiltonian path: a path that visits each verdex exactly once.

Graph Algorithms

a graph is called Hamiltonian if it has a Hamiltonian cycle.

Q_1: Input: a connected graph

Output: Is the graph Eulerian? Yes or NO

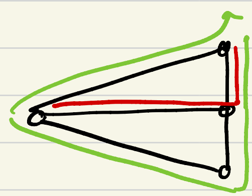
alg: check the degrees of all verdices.

If all degrees are even then output Yes, otherwise NO

Q_2: Input: a connected graph?

Output: Yes (If the graph is Hamiltonian), NO (otherwise)

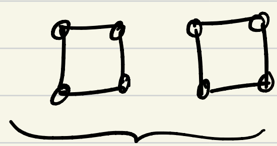
NP-herd



trivial: try all possible cycles (not efficient - can take for ever)

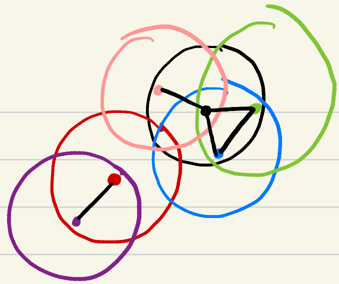
Q_1': Input: a graph?

Output: Is the graph is Eulerian?



Is the graph connected?

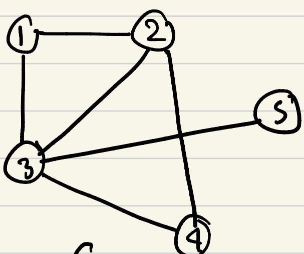
typpical algorithmic questions on graphs:



- Is G connected? ✓
- Is G bipardite? ✓
- Is G planar? Grad Courses
- Dos G have any bridges? ✓
- what is shortest path between two verdices? 3-4
  (shortest path algorithms)
- what is a minimum connected spaning subgraph? 3-4
  (minimum spaning tree)

⋮

Common way to represent a graph:
- adjacency matrix (arry)
- adjacency lists (linked lists)

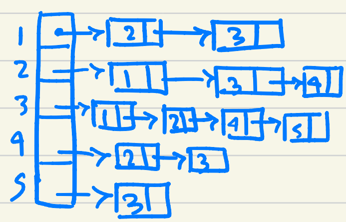


G

Matrix:

	1	2	3	4	5
1	0	1	1	0	0
2	1	0	1	1	0
3	1	1	0	1	1
4	0	1	1	0	0
5	0	0	1	0	0

lists:



Graph Serch Algorithms (visiting all vertices of graph)
- Breath First Serch (BFS): queue
- Depth First Serch (DFS): stack